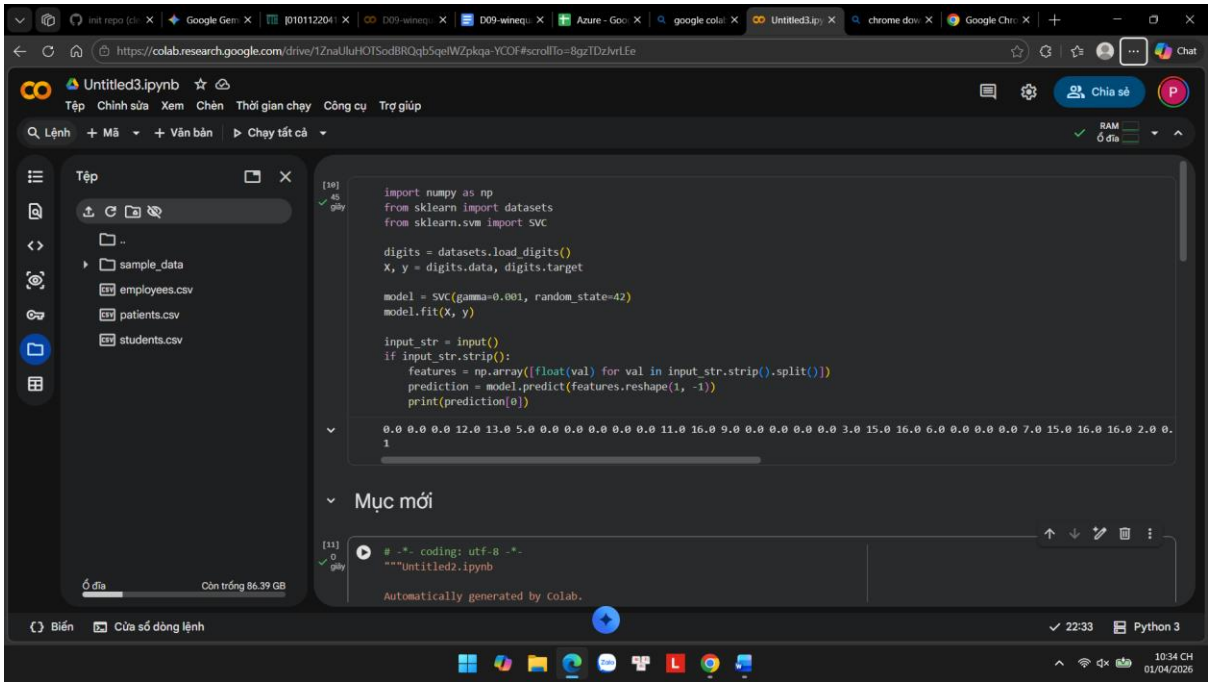
Lab 010426-1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Tệp Chỉnh sửa Xem Chèn Thời gian chạy Công cụ Trợ giúp
Lệnh + Mã + Văn bản Chạy tất cả
Chia sẻ
Tệp
sample_data
employees.csv
patients.csv
students.csv
Ổ đĩa Còn trống 86.39 GB
[10]
import numpy as np
from sklearn import datasets
from sklearn.svm import SVC
digits = datasets.load_digits()
X, y = digits.data, digits.target
model = SVC(gamma=0.001, random_state=42)
model.fit(X, y)
input_str = input()
if input_str.strip():
features = np.array([float(val) for val in input_str.strip().split()])
prediction = model.predict(features.reshape(1, -1))
print(prediction[0])
0.0 0.0 0.0 12.0 13.0 5.0 0.0 0.0 0.0 0.0 0.0 11.0 16.0 9.0 0.0 0.0 0.0 0.0 3.0 15.0 16.0 6.0 0.0 0.0 0.0 7.0 15.0 16.0 16.0 2.0 0.
1
Mục mới
[11]
-*- coding: utf-8 -*-
"""Untitled2.ipynb
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22:33
Python 3
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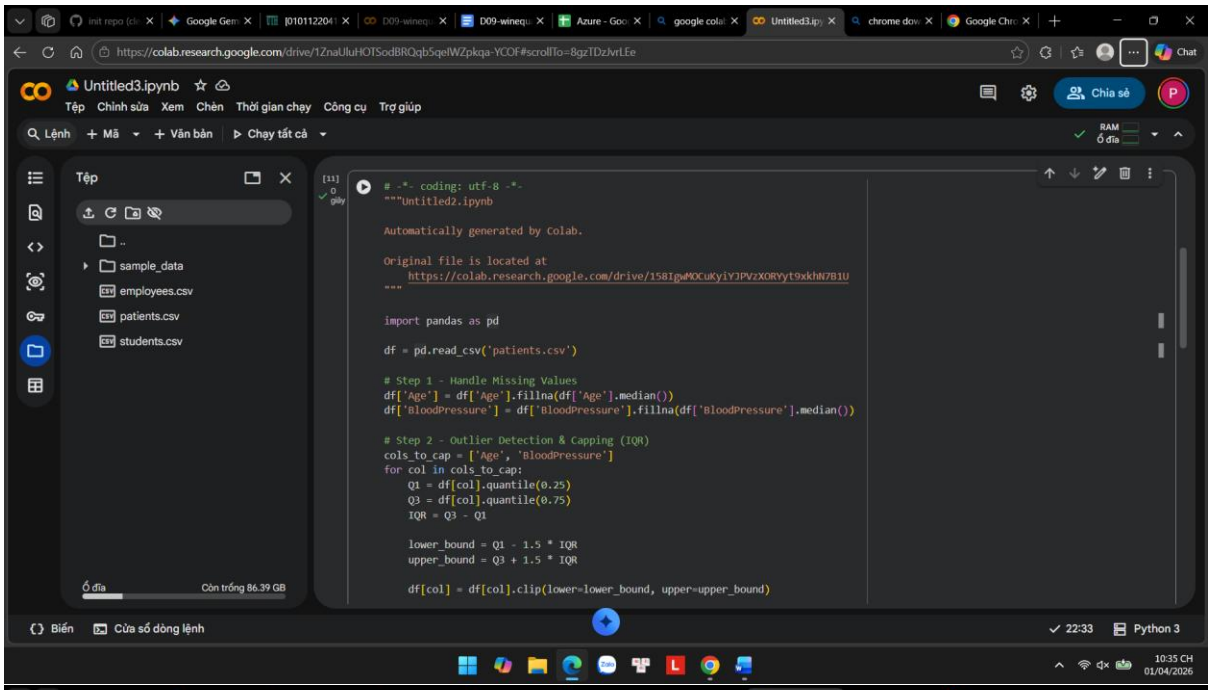
LabEL1-3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[11]
-*- coding: utf-8 -*-
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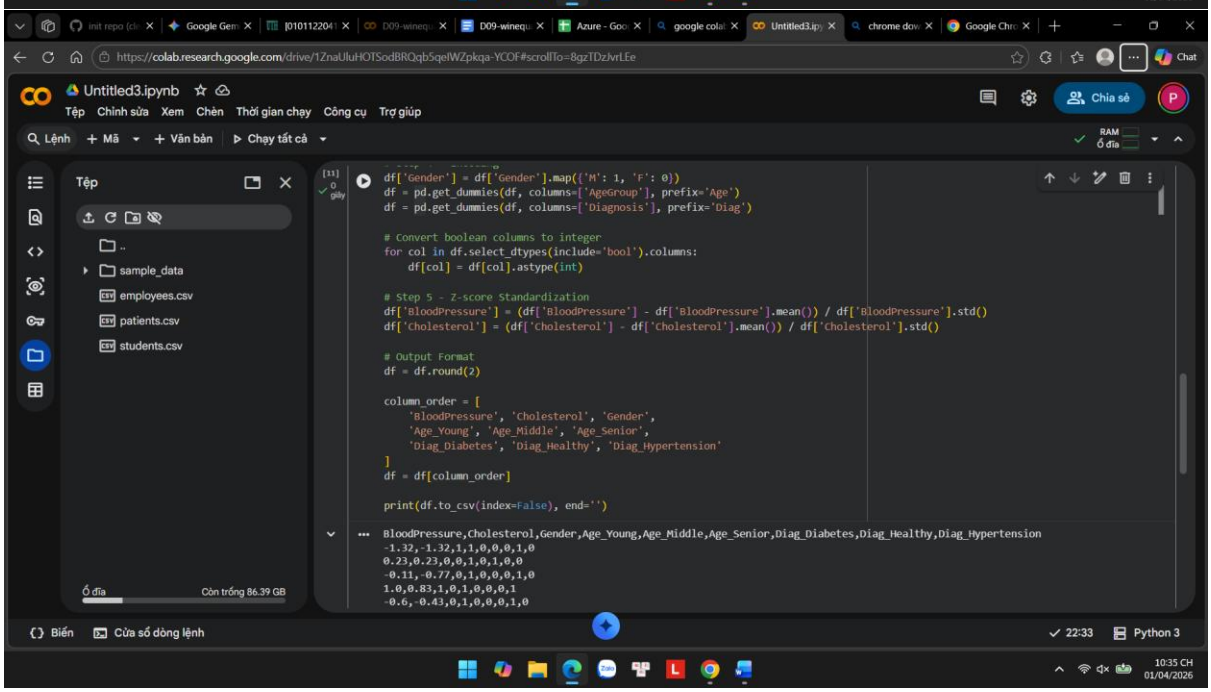
Automatically generated by Colab.
Original file is located at
https://colab.research.google.com/drive/158IgwMOCuKyiYJPVzXORYyt9xkhN7B1U
"""
import pandas as pd
df = pd.read_csv('patients.csv')
# Step 1 - Handle Missing Values
df['Age'] = df['Age'].fillna(df['Age'].median())
df['BloodPressure'] = df['BloodPressure'].fillna(df['BloodPressure'].median())
# Step 2 - Outlier Detection & Capping (IQR)
cols_to_cap = ['Age', 'BloodPressure']
for col in cols_to_cap:
Q1 = df[col].quantile(0.25)
Q3 = df[col].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
df[col] = df[col].clip(lower=lower_bound, upper=upper_bound)
Biến Cửa sổ dòng lệnh
22:33 Python 3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Ổ đĩa Còn trống 86.39 GB
[11]
df['Gender'] = df['Gender'].map({'M': 1, 'F': 0})
df = pd.get_dummies(df, columns=['AgeGroup'], prefix='Age')
df = pd.get_dummies(df, columns=['Diagnosis'], prefix='Diag')
# Convert boolean columns to integer
for col in df.select_dtypes(include='bool').columns:
df[col] = df[col].astype(int)
# Step 5 - Z-score Standardization
df['BloodPressure'] = (df['BloodPressure'] - df['BloodPressure'].mean()) / df['BloodPressure'].std()
df['Cholesterol'] = (df['Cholesterol'] - df['Cholesterol'].mean()) / df['Cholesterol'].std()
# Output Format
df = df.round(2)
column_order = [
'BloodPressure', 'Cholesterol', 'Gender',
'Age_Young', 'Age_Middle', 'Age_Senior',
'Diag_Diabetes', 'Diag_Healthy', 'Diag_Hypertension'
]
df = df[column_order]
print(df.to_csv(index=False), end='')
BloodPressure,Cholesterol,Gender,Age_Young,Age_Middle,Age_Senior,Diag_Diabetes,Diag_Healthy,Diag_Hypertension
-1.32,-1.32,1,1,0,0,0,1,0
0.23,0.23,0,0,1,0,1,0,0
-0.11,-0.77,0,1,0,0,0,1,0
1.0,0.83,1,0,1,0,0,0,1
-0.6,-0.43,0,1,0,0,0,1,0
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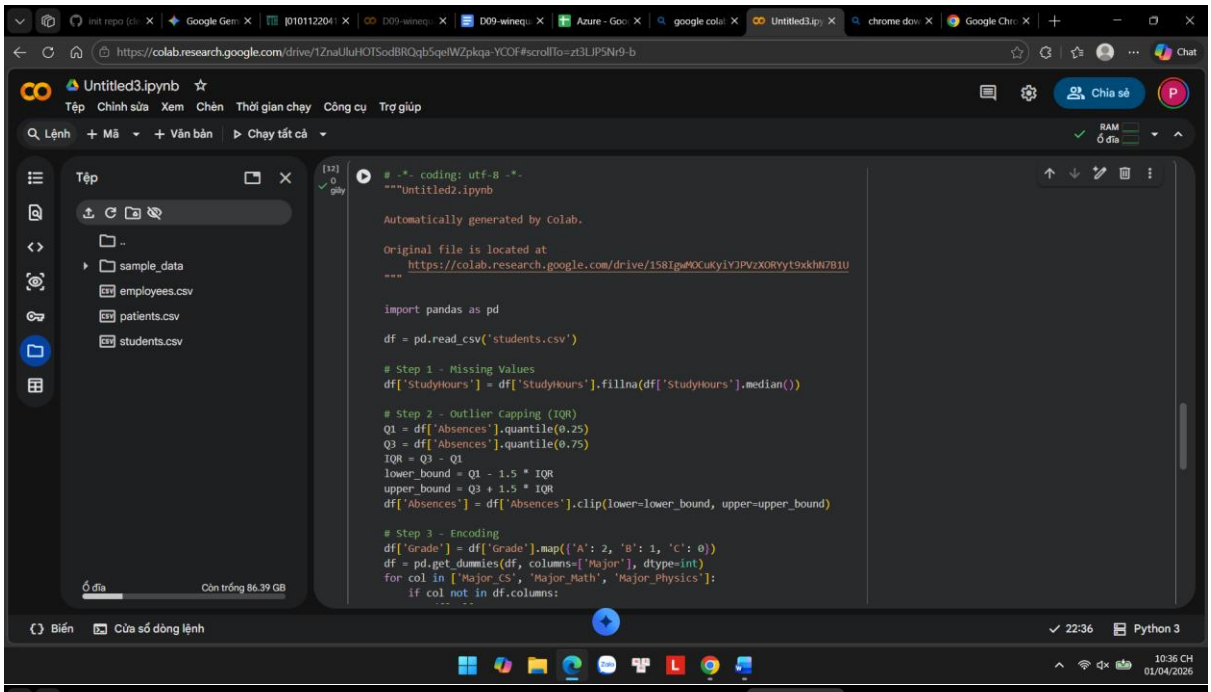
22:33 Python 3

labEL1-2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"""
import pandas as pd
df = pd.read_csv('students.csv')
# Step 1 - Missing Values
df['StudyHours'] = df['StudyHours'].fillna(df['StudyHours'].median())
# Step 2 - Outlier Capping (IQR)
Q1 = df['Absences'].quantile(0.25)
Q3 = df['Absences'].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
df['Absences'] = df['Absences'].clip(lower=lower_bound, upper=upper_bound)
Step 3 - Encoding
df['Grade'] = df['Grade'].map({'A': 2, 'B': 1, 'C': 0})
df = pd.get_dummies(df, columns=['Major'], dtype=int)
for col in ['Major_CS', 'Major_Math', 'Major_Physics']:
if col not in df.columns:
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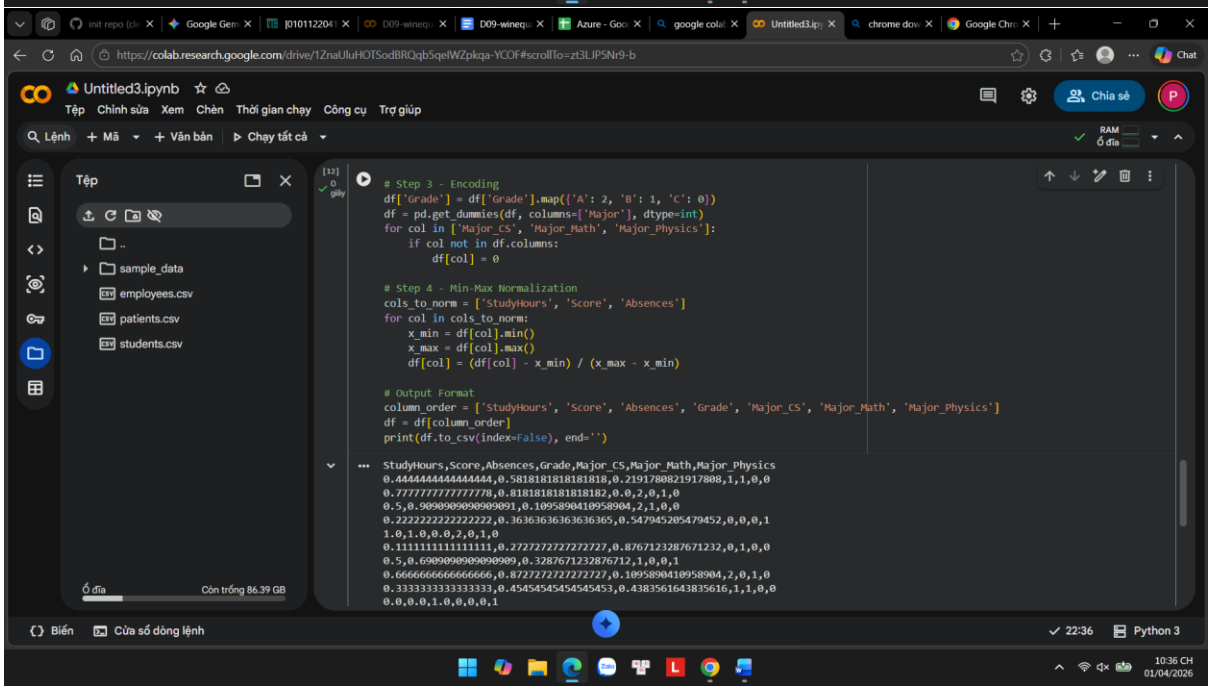
22:36 Python 3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Step 3 - Encoding
df['Grade'] = df['Grade'].map({'A': 2, 'B': 1, 'C': 0})
df = pd.get_dummies(df, columns=['Major'], dtype=int)
for col in ['Major_CS', 'Major_Math', 'Major_Physics']:
if col not in df.columns:
df[col] = 0
# Step 4 - Min-Max Normalization
cols_to_norm = ['StudyHours', 'Score', 'Absences']
for col in cols_to_norm:
x_min = df[col].min()
x_max = df[col].max()
df[col] = (df[col] - x_min) / (x_max - x_min)
# Output Format
column_order = ['StudyHours', 'Score', 'Absences', 'Grade', 'Major_CS', 'Major_Math', 'Major_Physics']
df = df[column_order]
print(df.to_csv(index=False), end='')
StudyHours,Score,Absences,Grade,Major_CS,Major_Math,Major_Physics
0.4444444444444444,0.5818181818181818,0.2191780821917808,1,1,0,0
0.7777777777777778,0.8181818181818182,0.0,2,0,1,0
0.5,0.9090909090909091,0.1095890410958904,2,1,0,0
0.2222222222222222,0.3636363636363636365,0.5479452054794520,0,0,0,1
1.0,1.0,0.0,2,0,1,0
0.1111111111111111,0.2727272727272727,0.8767123287671232,0,1,0,0
0.5,0.6909090909090909,0.3287671232876712,1,0,0,1
0.6666666666666666,0.8727272727272727,0.1095890410958904,2,0,1,0
0.3333333333333333,0.4545454545454545453,0.4383561643835616,1,1,0,0
0.0,0.0,1.0,0,0,0,1
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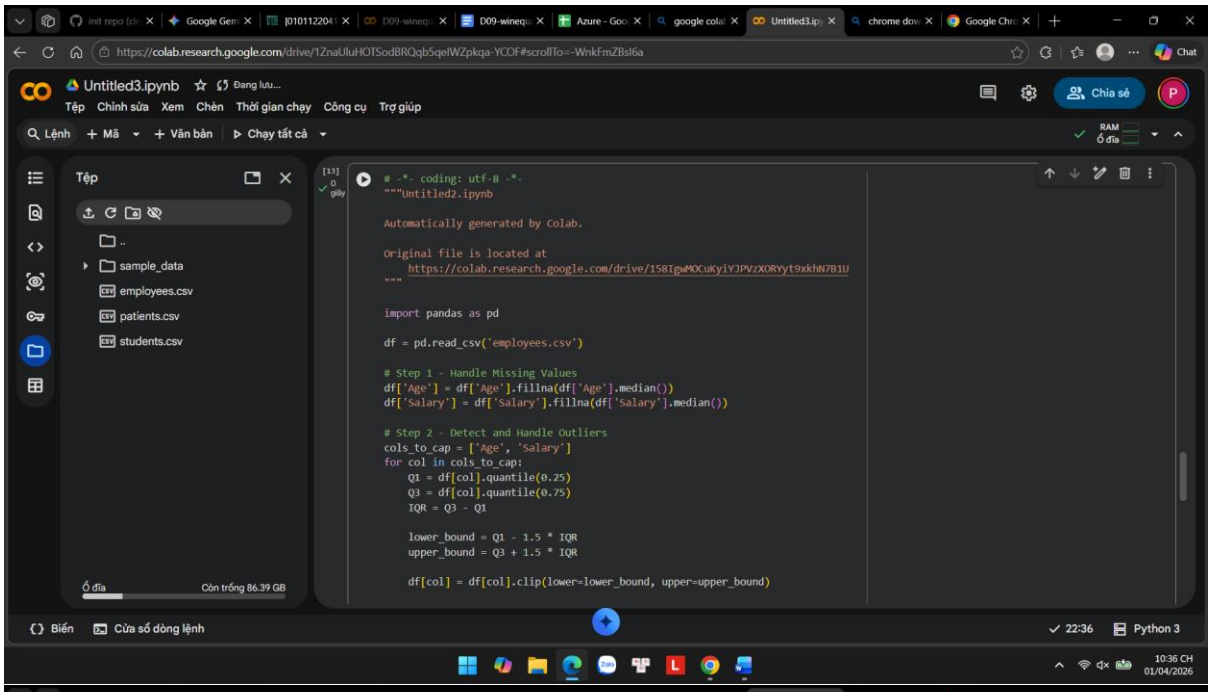
labEL1-1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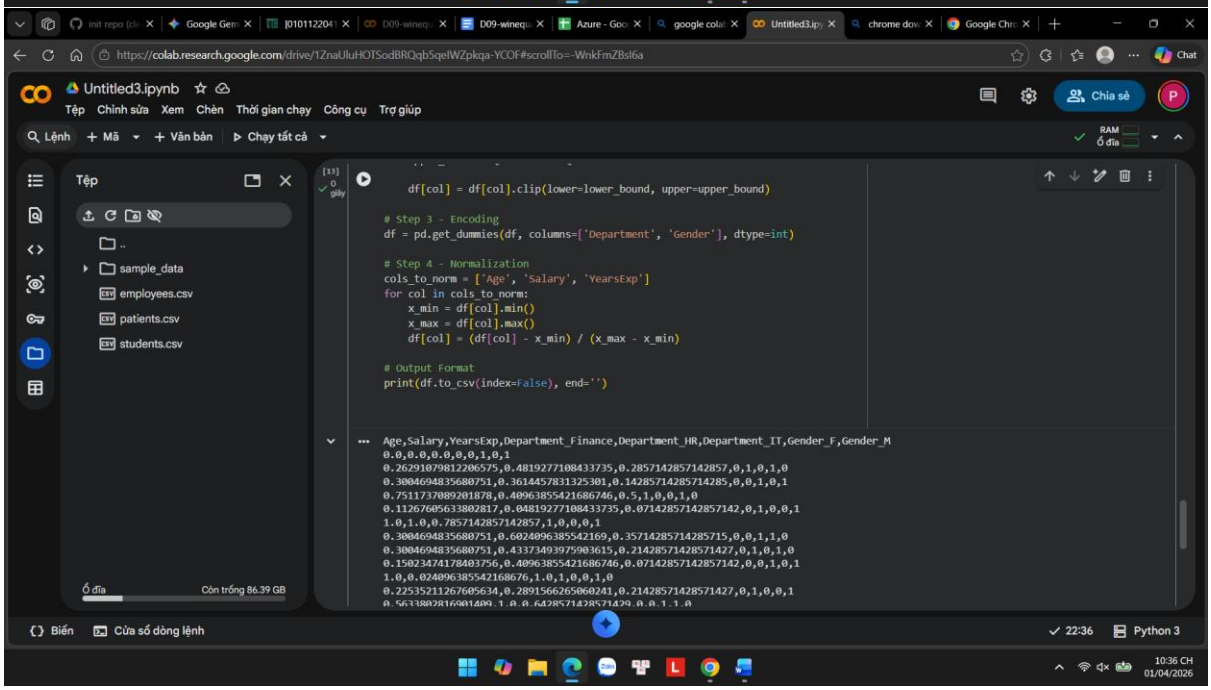
"""
import pandas as pd
df = pd.read_csv('employees.csv')
# Step 1 - Handle Missing Values
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Salary'] = df['Salary'].fillna(df['Salary'].median())
# Step 2 - Detect and Handle Outliers
cols_to_cap = ['Age', 'Salary']
for col in cols_to_cap:
Q1 = df[col].quantile(0.25)
Q3 = df[col].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
df[col] = df[col].clip(lower=lower_bound, upper=upper_bound)
Biến Cửa sổ dòng lệnh
22:36 Python 3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df[col] = df[col].clip(lower=lower_bound, upper=upper_bound)
# Step 3 - Encoding
df = pd.get_dummies(df, columns=['Department', 'Gender'], dtype=int)
# Step 4 - Normalization
cols_to_norm = ['Age', 'Salary', 'YearsExp']
for col in cols_to_norm:
x_min = df[col].min()
x_max = df[col].max()
df[col] = (df[col] - x_min) / (x_max - x_min)
# Output Format
print(df.to_csv(index=False), end='')
Age,Salary,YearsExp,Department_Finance,Department_HR,Department_IT,Gender_F,Gender_M
0.0,0.0,0.0,0,0,1,0,1
0.2629107981220657 5,0.4819277108433735,0.2857142857142857,0,1,0,1,0
0.3004694835680751,0.3614457831325301,0.1428571428571428 5,0,0,1,0,1
0.7511737089201878,0.4096385542168674 6,0.5,1,0,0,1,0
0.1126760563380281 7,0.0481927710843373 5,0.0714285714285714 2,0,1,0,0,1
1.0,1.0,0.7857142857142857,1,0,0,0,1
0.3004694835680751,0.6024096385542169,0.3571428571428571 5,0,0,1,1,0
0.3004694835680751,0.4337349397590361 5,0.2142857142857142 7,0,1,0,1,0
0.1502347417840375 6,0.4096385542168674 6,0.0714285714285714 2,0,0,1,0,1
1.0,0.0240963855421686 76,1.0,1,0,0,1,0
0.2253521126760563 4,0.2891566265060241,0.2142857142857142 7,0,1,0,0,1
Biến Cửa sổ dòng lệnh
22:36 Python 3